

1D Convolutional Autoencoder for ECG Signal Denoising and Classification

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ABSTRACT

INDEX TERMS CAE, Classification, CNN, CVD, Decoder, Denoising, DWT, ECG, EMD, Encoder, MLII, SWT

I. INTRODUCTION

Cardiovascular disease (CVD) is one of the leading causes of death globally, and it involves all disorders that affect the heart and blood circulation or vessels [1]. For this reason, it is necessary to enhance all detection techniques to be able to early detect any irregularities to avoid fatal stages.

One of the most important techniques is the electrocardiogram (ECG), which is a test that can be used to monitor a patient's heart rhythm and electrical activity. The electrical signals produced by the heart are measured using electrodes attached to the patient's skin. The heart activity is recorded to be analyzed by specialists to investigate symptoms of possible heart problems or any abnormalities.

Despite the usefulness of ECG, the recorded signals are prone to various noise sources, which may distort the heart's signal, leading to misdiagnosis. There are many reasons for having noise during measurement.

The following are the main causes of noise:

- Noises caused by electrode movement (EM), which usually have a large amplitude.
- High-Frequency Noises caused by muscle artifact (MA), which is a muscle noise caused by involuntary muscle movements like shivering, tension, or tremors.
- Low-Frequency Noises caused by baseline wander (BW), mainly caused by human breathing /respiration [2].

The mentioned noise types can mask important diagnostic features in the signal, so effectively removing these noises without altering the essential morphology of the ECG signal remains a significant and ongoing challenge[3].

Thus, various denoising techniques have been developed to enhance the quality of ECG signals. This study aims to implement and train a neural network-based filtering/denoising technique using a convolutional autoencoder(CAE).

This paper is structured as follows:

- **Section II** which covers the necessary background information, starting from defining the ECG signal and its characteristics, such as the QRS complex, and waves within ECG cycle like P-waves, and T-waves, then moves to a brief about classical ECG signal denoising techniques, and then introduces the necessity of data-driven techniques, finally it illustrates the needed technologies in this paper like 1D Convolution and Autoencoder.
- **Section III** which outlines
- **Section IV** which shows
- **Section VI** which compares
- Lastly, **Section VII** explains the major challenges and research gaps, showing that denoising the ECG signal is a very active topic in research, and there are multiple future directions that need to be explored.

II. BACKGROUND AND LITERATURE REVIEW

A. ECG SIGNAL AND ITS CHARACTERISTICS

A full ECG is obtained via 10 electrodes, that captures 12 leads (signals). Each lead identifies the electrical activity from a different angle/perspective in order to get a complete picture of the heart. However, a single lead can offer adequate information for a quick and initial assessment of the patient (Modified Limb Lead 2 (MLII) is used in this paper, which

provides a clear QRS complex and visible P-Waves making it suitable for heartbeat detection). One ECG cycle consists of five waves: P, Q, R, S, and T that map different phases of the heart's activities.

The P-wave is the first positive deflection on the ECG, The QRS represents the simultaneous activation of the right and left ventricles, although most of the QRS waveform is derived from the larger left ventricular musculature and the T-wave should be concordant with the QRS complex. P-R interval measured from the beginning of the P-wave to the first deflection of the QRS complex and Q-T interval measured from the first deflection of QRS complex to end of T-wave at the isoelectric line as shown in Fig. 1 [4].

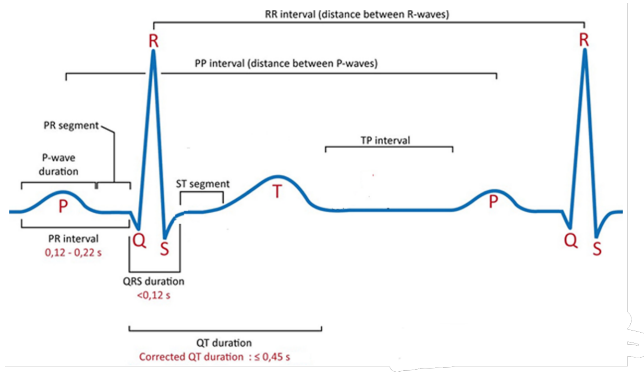


FIGURE 1. ECG Waveform [4]

The nominal values for the mentioned intervals in Lead II are as shown in Table 1.

TABLE 1. Characteristics of normal ECG waves[4]

Waves and intervals	Duration	Amplitude
P-waves	< 0.11 s	< 2.5 mm
QRS-complex	< 0.12 s	> 0.5 mV
T-waves	N/A	N/A
P-R interval	0.12–0.22 s	N/A
S-T interval	N/A	> 0.5 mm
Q-T interval	< 0.45 s	N/A

B. TRADITIONAL ECG SIGNAL DENOISING TECHNIQUES

There are several techniques that addressed ECG signal denoising, including **time-frequency analysis techniques** such as wavelet transforms like discrete wavelet transform(DWT), and Stationary wavelet transform (SWT), and techniques like **empirical mode decomposition (EMD)** and **filtering-based techniques**, which are widely used due to their effectiveness in ECG denoising [5]. These techniques have shown significant results in separating noise from the ECG signal. For instance, wavelet transform-based techniques are efficient at isolating components in both time and frequency domains to remove baseline drift or high-frequency noise. On the other hand, adaptive filters can flexibly adjust to variable noise

conditions, while EMD decomposes signals into intrinsic mode functions (IMF) to isolate noise-dominant components [6], [7]. Despite this, such classical techniques are not sufficient as they face notable challenges and limitations that struggle with the critical morphological characteristics of the ECG signal. For instance, muscle artifact noise often overlaps with the frequency bands of P-waves, T-waves, and QRS complexes (important features of ECG signals), leading to signal distortion or loss of clinically relevant information. These overlaps make it difficult for customized filters to distinguish between noise and meaningful signal content, especially under low SNR conditions [8], [9].

C. DATA-DRIVEN ECG SIGNAL DENOISING TECHNIQUES

D. 1-D CONVOLUTION

1-D Convolution is a fundamental signal processing algorithm in deep learning that is widely used with ECG signals. 1D-CNNs are designed to handle one-dimensional data, such as time-series data, sequences like text or any data where the primary structure is along a single axis. The kernel (or filter) in a 1-D CNN moves along one dimension. If the data is represented as a vector $[x_1, x_2, \dots, x_n]$, the kernel will slide over this vector to detect patterns within the sequence. The shape of the kernel is a 1-D array with dimension $(k,)$, where k is the size of the kernel. The kernel slides along the single axis of the input vector. For example, in a time-series application, the kernel moves along the time axis to capture temporal patterns. The receptive fields of a 1-D CNN kernel is a contiguous segment of the input sequence. Therefore, as the kernel slides over the input, it aggregates information from k consecutive elements. For an input sequence x and a kernel w , the convolution operation in

$$(x * w)(t) = \sum_{i=0}^{k-1} x(t+i) \cdot w(i)$$

where:

- x is the input sequence,
- w is the kernel (or filter),
- $(x * w)(t)$ denotes the convolution of x and w at position t
- k is the size of the kernel,
- $x(t+i)$ is the element of the input sequence at position $t+i$,
- $w(i)$ is the element of the kernel at position i .

E. AUTOENCODER

III. METHODOLOGY

A. DATASET

The MIT-BIH arrhythmia database [12] was adopted in the paper. It contains 48 slightly more than 30-minute records obtained from 47 subjects/patients and extracted from two leads (lead II (MLII) and lead V1). The subjects(47) are divided into 22 women of an age range from 23 to 90 years

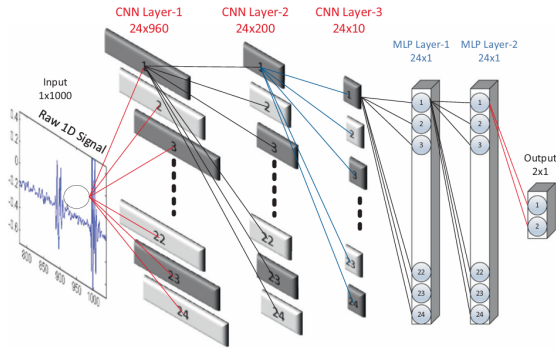


FIGURE 2. 1-D CNN Sample [10]

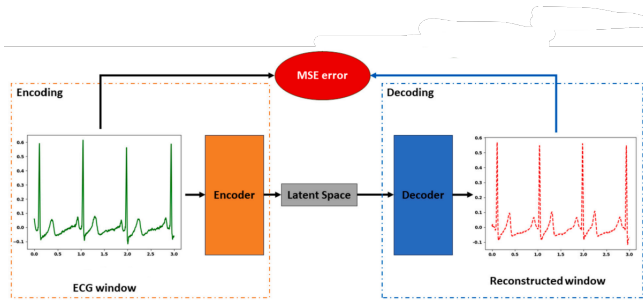


FIGURE 3. Autoencoder Architecture [11]

and 25 men of an age range from 32 to 90 years. The recordings were digitized at 360 samples per second per channel with 11-bit resolution over a 10-mV range.

MIT-BIH arrhythmia database is divided into two classes: normal 25 ECG records and abnormal 23 ECG records. In this paper, the algorithm is tested on the ECG signals taken from lead II (MLII).

MIT-BIH arrhythmia database is a standard benchmark in the ECG denoising literature, enabling direct comparison with published results.

Table 2 shows the train, validation, and test splits of samples:

TABLE 2. Division of MIT-BIH arrhythmia samples

Set	Number of Records
Training	22155
Validation	3165
Testing	3165

For noisy signals generation, MIT-BIH Noise Stress Test Database (NSTDB) [13] has been used, which contains noise recordings of the three dominant ECG artifact types; Baseline Wander (BW), Muscle Artifact (MA), and Electrode Motion (EM). Noise is added randomly from Baseline Wander (BW) and Muscle Artifact (MA). Noisy signals contain only one noise type to simplify the model, and Electrode motion (EM) artifact is not added, as it is generally considered the most troublesome, since it can mimic the appearance of ectopic beats and cannot be removed easily by simple filters, as can

noise of other types.

B. ARCHITECTURE

IV. EVALUATION

A. EVALUATION METRICS

V. RESULTS

VI. ABLATION STUDIES

VII. CHALLENGES AND FUTURE DIRECTIONS

VIII. CONCLUSION

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